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# Software Requirement Specification (SRS)

PROJECT: Security incident mapping (SIM)

## 1.0 Introduction:

The Security Incident Mapping (SIM) system is a web-based application designed to provide an e icient platform for reporting, tracking, and managing security incidents. The system enables students, and landlords to submit incident reports, identify crime-prone locations, and assist security personnel in responding promptly to emergencies. By maintaining a centralized database of reported incidents and presenting them in an organized manner, the SIM system supports better decisionmaking, enhances communication among stakeholders, and contributes to creating a safer and more secure campus environment. This Software Requirements Specification (SRS) document defines the functional and non-functional requirements, system features, user interactions, and other specifications necessary for the successful development and implementation of the Security Incident Mapping system.

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the Security Incident Mapping (SIM) system. It serves as a comprehensive guide for the design, development, implementation, testing, and maintenance of the proposed web-based application. The document clearly describes the objectives of the system, the features it will provide, the expected behavior of the software, and the interactions between users and the system. It also establishes a common understanding among developers, project supervisors, testers, and stakeholders regarding the scope and expected functionality of the application.

Furthermore, this SRS is intended to ensure that the developed system e ectively addresses the security challenges experienced within the campus environment by providing a centralized platform for reporting, managing, and monitoring security incidents. It specifies the requirements necessary to build a secure, reliable, and userfriendly application that supports students, sta , landlords, security o icers, and administrators in carrying out their respective roles. By documenting these requirements before implementation, the SRS helps reduce ambiguity, improve communication among project stakeholders, and ensure that the final system meets the intended objectives and user needs.

## 1.2 Scope

The scope of the Security Incident Mapping (SIM) system covers the design and development of a web-based application that facilitates the reporting, monitoring, and management of security incidents within and around the university campus. The system is intended for use by students, sta , landlords, security o icers, and system administrators. It provides users with the ability to register, log in, report security incidents, submit relevant details such as the type of incident, location, date, and description, and, where applicable, upload supporting evidence. The system also enables users to view security information and receive updates regarding reported incidents, thereby improving communication between the campus community and security personnel.

In addition, the SIM system provides administrative features that allow authorized personnel to review, verify, manage, and update reported incidents through a centralized dashboard. It supports the identification of crime-prone areas by displaying incidents on a map, generating reports, and maintaining a secure database of incident records for future reference and analysis. The system is designed to operate on standard web browsers and is accessible through internet-enabled devices such as smartphones, tablets, and computers. However, the scope of this project is limited to the development of the web application and its core functionalities. It does not include the deployment of physical security devices such as CCTV cameras, alarm systems, or GPS hardware, but it is designed to support future integration with such technologies if required.

## 1.3 Users

The Security Incident Mapping (SIM) system is designed to serve di erent categories of users within the university community, with each user having specific roles and access privileges. The primary users include students, sta , landlords, security o icers, and the system administrator. Students, sta , and landlords are responsible for reporting security incidents by providing accurate information such as the type of incident, location, date, time, and a brief description. They can also log into the system to view the status of their submitted reports and receive updates on reported incidents.

Security o icers and system administrators are responsible for managing and monitoring the system. They review and verify submitted incident reports, update the status of ongoing cases, identify crime hotspots through incident mapping, and generate reports for security planning and decision-making. The administrator also manages user accounts, maintains the system database, and ensures that the application functions e iciently and securely. By assigning di erent responsibilities to each user category, the SIM system ensures e ective collaboration and promotes a safer campus environment.

## 2.0 Functional Requirements

The functional requirements describe the specific functions and services that the Security Incident Mapping (SIM) system must perform to meet the needs of its users. The system shall provide an interactive and user-friendly platform that enables students, sta , landlords, security o icers, and administrators to perform their respective tasks e iciently. These requirements define the expected behavior of the system and ensure that all essential operations are carried out correctly.

The SIM system shall support user registration and authentication to ensure that only authorized users can access secured features of the application. It shall allow users to report security incidents by entering details such as the incident type, location, date, time, and description, with the option of attaching supporting evidence where necessary. The system shall display reported incidents on a map to identify crime-prone areas, enable users to view the status of their reports, and provide notifications when updates are made. Furthermore, the system shall allow administrators and security o icers to review, verify, edit, and manage reported incidents, generate security reports, manage user accounts, and maintain a centralized database of all reported security incidents for e ective monitoring and decision-making.

Functional Requirements List

- FR1: The system shall allow users to register an account.
- FR2: The system shall allow registered users to log in and log out securely.
- FR3: The system shall allow users to report security incidents.
- FR4: The system shall allow users to provide the incident location, date, time, and description.
- FR5: The system shall display reported incidents on a crime map.
- FR6: The system shall allow users to view the status of submitted reports.
- FR7: The system shall notify administrators when a new incident is reported.
- FR8: The system shall allow administrators to review, update, and manage incident reports.
- FR9: The system shall maintain a secure database of all reported incidents.
- FR10: The system shall generate reports and statistics to support security planning and decision-making.

## 3.0 Non-Functional Requirements

The non-functional requirements describe the quality attributes and performance standards that the Security Incident Mapping (SIM) system must satisfy to ensure e icient, secure, and reliable operation. Unlike functional requirements, which specify what the system should do, non-functional requirements define how well the system should perform. These requirements focus on aspects such as usability, reliability, security, performance, availability, maintainability, and compatibility to ensure that the system provides a satisfactory user experience and operates e ectively under normal conditions.

The SIM system shall be designed with a simple and user-friendly interface that enables users with little or no technical knowledge to navigate the application easily. The system shall provide fast response times when users submit reports or retrieve information and shall protect user data through secure authentication and access control mechanisms. It shall also be reliable enough to store incident records accurately without data loss and remain available whenever users need to report emergencies. Furthermore, the system shall be easy to maintain, support future upgrades, and operate e iciently across di erent web browsers and internet-enabled devices.

Non-Functional Requirements List

- NFR1 – Performance: The system shall process user requests and display results within a few seconds under normal operating conditions.
- NFR2 – Security: The system shall require user authentication, protect user information, and restrict administrative functions to authorized users only.
- NFR3 – Reliability: The system shall store and retrieve incident data accurately without loss or corruption.
- NFR4 – Usability: The interface shall be simple, intuitive, and easy for students, sta , landlords, and security personnel to use.
- NFR5 – Availability: The system shall be available to users 24 hours a day, 7 days a week, except during scheduled maintenance.
- NFR6 – Maintainability: The system shall be designed to allow easy updates, bug fixes, and future enhancements.
- NFR7 – Compatibility: The system shall function correctly on modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and on desktop and mobile devices.
- NFR8 – Scalability: The system shall support an increasing number of users and incident reports without significant performance degradation.

## 4.0 Hardware Requirements

The Security Incident Mapping (SIM) system requires basic computer hardware to ensure smooth operation and accessibility for all users. Since the system is web-based, users can access it through internet-enabled devices such as desktop computers, laptops, tablets, or smartphones. The hardware should be capable of running a modern web browser and maintaining a stable internet connection for submitting incident reports, viewing security information, and accessing system features. Security o icers and administrators may use computers with higher processing capabilities to e iciently manage incident records, monitor reports, and perform administrative tasks.

The server hosting the application should have adequate processing power, memory, and storage capacity to support multiple users simultaneously and ensure reliable data storage. A stable network connection is also essential to facilitate uninterrupted communication between users and the system. The recommended hardware components are listed below.

Hardware Requirements List

Client Device: Desktop computer, laptop, tablet, or smartphone.

Processor: Intel Core i3 (or equivalent) and above.

Memory (RAM): Minimum 4 GB (8 GB recommended).

Storage: Minimum 20 GB of available disk space.

Display: Monitor with a minimum resolution of 1366 × 768 pixels.

Internet Connection: Stable broadband or mobile internet connection.

Server (if deployed): Multi-core processor, minimum 8 GB RAM, adequate storage, and reliable network connectivity.

## 5.0 Software Requirements

The Security Incident Mapping (SIM) system is designed as a web-based application and therefore requires both development and operational software to function e ectively.

During development, programming languages, web technologies, and development tools are required to build, test, and maintain the application. During deployment and usage, users only need a compatible web browser and a stable internet connection to access the system. The selected software components should be reliable, easy to maintain, and capable of supporting future enhancements.

The system will be developed using HTML and CSS for designing the user interface, while JavaScript may be used to add interactivity and improve user experience. The application can be tested locally using XAMPP or any suitable local web server before deployment. Source code management will be carried out using GitHub to ensure proper version control and collaboration. The system is expected to run on modern operating systems such as Windows, Linux, and macOS, and should be accessible through standard web browsers without requiring users to install additional software.

Software Requirements List

Operating System: Windows 10/11, Linux, or macOS.

Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.

Front-end Technologies: HTML5 and CSS3.

Programming Language (Optional): JavaScript (for interactive features).

Development Environment: Visual Studio Code (VS Code) or any suitable code editor.

Local Web Server (for testing): XAMPP or Live Server Extension.

Version Control: Git and GitHub.

Internet Connection: Required for accessing the web application and synchronizing project files.

## 6.0 Assumptions

The development of the Security Incident Mapping (SIM) system is based on several assumptions to ensure that the application operates effectively and achieves its intended objectives. It is assumed that all users, including students, staff, landlords, security officers, and administrators, have access to internet-enabled devices such as smartphones, tablets, or computers, and possess basic knowledge of how to use a web-based application. It is also assumed that users will provide accurate, truthful, and complete information when reporting security incidents to ensure the reliability of the data stored in the system.

Furthermore, it is assumed that the university management and security personnel will actively monitor the system, verify reported incidents, and take appropriate action when necessary. The system also assumes the availability of a stable internet connection and adequate server resources to support continuous access and reliable data storage. Finally, it is assumed that authorized administrators will properly maintain the system, manage user accounts, and perform regular updates to ensure the security, reliability, and e ective operation of the Security Incident Mapping (SIM) system.

## 8.0 Constraints

The development and operation of the Security Incident Mapping (SIM) system are subject to certain limitations that may a ect its performance and implementation. Since the application is web-based, it requires a stable internet connection for users to access its features and submit incident reports. Users without internet connectivity or compatible devices may be unable to use the system e ectively. In addition, the accuracy of incident mapping depends largely on the correctness of the information provided by users. Incorrect or incomplete details may result in inaccurate reports and reduce the e ectiveness of the system in identifying security hotspots.

Furthermore, the system is limited by available hardware, software, and financial resources allocated for its development and maintenance. Only authorized administrators and security o icers are permitted to access administrative functions such as managing user accounts, updating incident records, and generating reports. The system also relies on regular maintenance, database backups, and software updates to ensure continuous operation and security. Although the SIM system enhances the reporting and management of security incidents, it does not replace the responsibilities of campus security personnel or guarantee the complete prevention of criminal activities within the campus environment.

## Constraints List

- A stable internet connection is required to access the system.
- Users must have an internet-enabled device and a compatible web browser.
- The accuracy of reports depends on the information provided by users.
- Administrative functions are restricted to authorized personnel.
- The system requires regular maintenance and database backups.
- Limited financial and technical resources may a ect future upgrades and expansion.

## 9.0 Conclusion

The Security Incident Mapping (SIM) system provides an e ective and reliable solution for improving the reporting, monitoring, and management of security incidents within the university community. By replacing traditional manual reporting methods with a centralized web-based platform, the system enables students, sta , landlords, security o icers, and administrators to communicate more e iciently and respond to security issues in a timely manner. The integration of incident reporting, data management, and location-based mapping enhances the ability to identify crime-prone areas, improve decision-making, and support proactive security measures across the campus.

In conclusion, the successful implementation of the SIM system will contribute significantly to creating a safer and more secure learning environment. The system promotes accurate record-keeping, faster emergency response, improved collaboration among stakeholders, and better security planning through the analysis of reported incidents. Although the system has certain limitations, its benefits outweigh its constraints and provide a strong foundation for future enhancements such as mobile application support, GPS integration, real-time notifications, and advanced data analytics. Overall, the Security Incident Mapping (SIM) system is expected to improve campus safety, strengthen security management, and enhance the overall well-being of the university community.